

REINFORCEMENT LEARNING

Problem: No ideal output value in training data

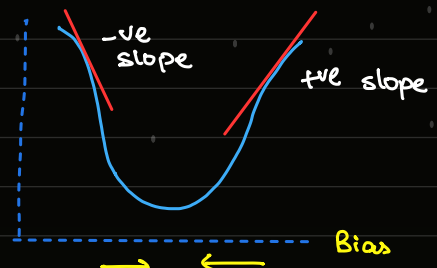
policy gradient

→ guess

correct guess → +ve reward

incorrect guess → -ve reward

Back Propagation



$$\text{derivative} * \text{reward} = \text{updated derivative}$$

to modify Bias correctly

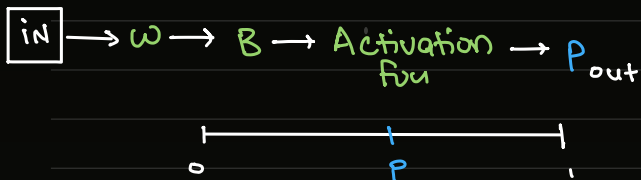
example:- $0.5 * 1$ (correct guess) = 0.5 → No change in derivative

$0.5 * (-1)$ [wrong guess] = -0.5 → inverse the slope

Step size = learning rate * updated derivative

New Bias = old bias - step size

Cross Entropy → $-\ln(\text{probability})$



Chose a number b/w 0 & 1

Do that action/output

Guess the output that should have come → P_{guess}

P_{guess} and P_{out} are connected by a line. Below this line, it says 'Cross Entropy → relation b/w guess and output to update bias'. To the right, it says 'if guess is correct then output will get closer to guess'.

$$\frac{d \text{ Cross Entropy}}{d \text{ Bias}}$$

$$\text{Cross Entropy} = -\log(P_{\text{out}})$$

Now we need to check if guess was correct → give Reward
if guess is wrong, negative reward will flip direction of derivative, to correct direction of training

$$\frac{d \text{ Cross Entropy}}{d \text{ Bias}} \times \text{Reward}$$

New Bias =

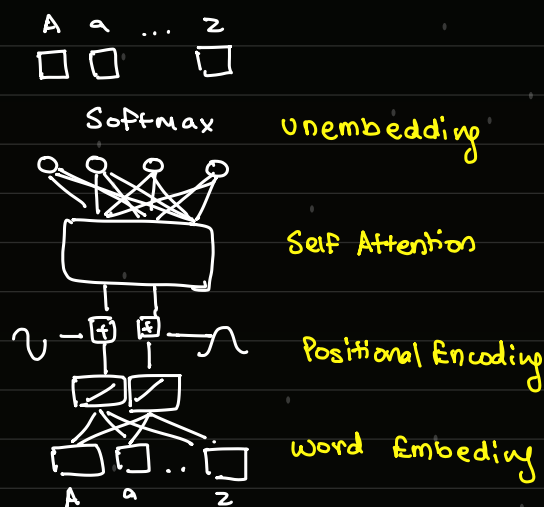
$$\text{old Bias} - \left(\text{learning rate} \times \frac{d \text{ C.E} \times \text{Reward}}{d \text{ Bias}} \right)$$

RLHF

How to train LLM

Pre Training:- Training to predict next word with huge text

example:- Hello, my name is



Aligning A Pre-Trained model

- ① Supervised fine Tuning
- ② RLHF

Supervised fine tuning :- Human made prompts + Human Responses

untrained, decoder only transformer
↳ wt, bias not set

Now model can run on the prompts it was trained for, but not new ones

This makes model overfit

RLHF: create large dataset while minimizing cost

SFT > RLHF but
is very expensive

use the output of softmax as probabilities that each token is selected

use that probability to select output tokens to get different answers each time

Create a list of pairs of responses and ask people which one they prefer

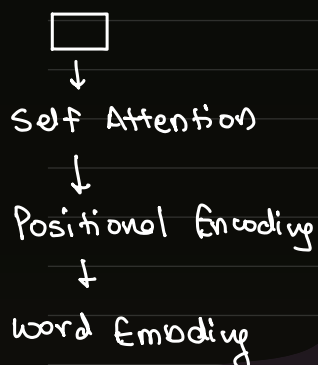
ex

AI is good ✗

AI is great ✓

Reinforcement Learning with human feedback

Process → Remove unembedding layer from model which is trained on Supervised fine tuning, replace with single output



give +ve reward and -ve reward [model can do this on own]

Pass both prompt and response on new model that calculates reward

example → OPENAI $-1 \left(\log \left(\text{sigmoid} \left(\text{Reward}_{\text{good}} - \text{Reward}_{\text{bad}} \right) \right) \right)$